

HARDWARE TROUBLESHOOTING AND CAPSTONE INTRODUCTION

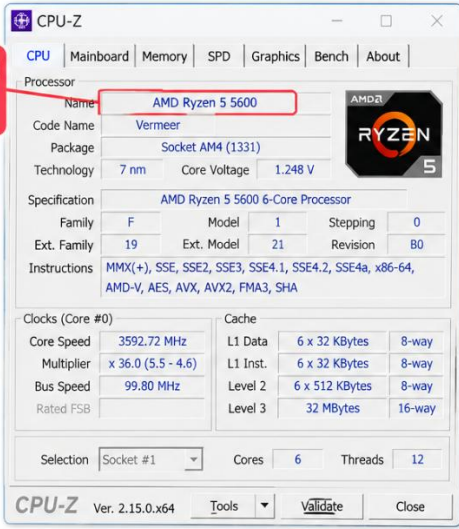
1. Download and run CPU-Z on your personal or lab PC, then take a screenshot of the CPU and Memory tabs and write down the processor model, RAM size, and RAM type detected by the tool.

->



CPU-Z CPU and Memory Tabs

EXAMPLE — replace with your PC readings



Processor model (look at the Name field)

Processor Name: AMD Ryzen 5 5600

Code Name: Vermeer

Package: Socket AM4 (1331)

Technology: 7 nm

Core Voltage: 1.248 V

Specification: AMD Ryzen 5 5600 6-Core Processor

Family: F

Ext. Family: 19

Ext. Model: 21

Revision: 80

Instructions: MMX(+), SSE, SSE2, SSE3, SSE4.1, SSE4.2, SSE4a, x86-64, AMD-V, AES, AVX, AVX2, FMA3, SHA

Clocks (Core #0)

Core Speed: 3592.72 MHz

Multiplier: x 36.0 (5.5 - 4.6)

Bus Speed: 99.80 MHz

Rated FSB:

Cache

L1 Data: 6 x 32 KBytes 8-way

L1 Inst: 6 x 32 KBytes 8-way

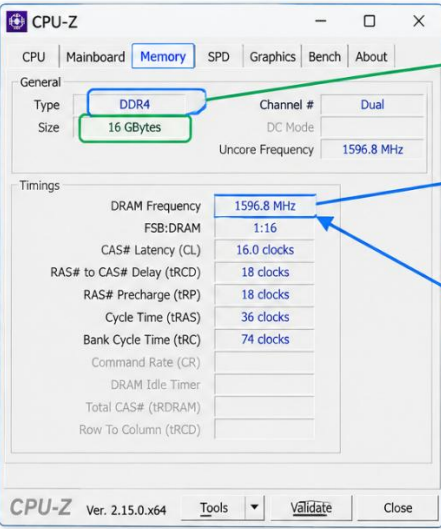
Level 2: 6 x 512 KBytes 8-way

Level 3: 32 MBytes 16-way

Selection: Socket #1

Cores: 6

Threads: 12



RAM size (look at the Size field)

RAM type (look at the Type field)

DRAM Frequency (helps identify the RAM type)

General

Type: DDR4

Size: 16 GBytes

Channel #: Dual

DC Mode:

Uncore Frequency: 1596.8 MHz

Timings

DRAM Frequency: 1596.8 MHz

FSB:DRAM: 1:16

CAS# Latency (CL): 16.0 clocks

RAS# to CAS# Delay (tRCD): 18 clocks

RAS# Precharge (tRP): 18 clocks

Cycle Time (tRAS): 36 clocks

Bank Cycle Time (tRC): 74 clocks

Command Rate (CR):

DRAM Idle Timer:

Total CAS# (tRDRAM):

Row To Column (tRCD):



Key Information (from the example above)

Processor Model	RAM Size	RAM Type
AMD Ryzen 5 5600	16 GBytes	DDR4

CPU-Z Verification

1. Download **CPU-Z** from the official **CPUID** website. The current Windows version listed by CPUID is **CPU-Z 3.01**, released August 20, 2026. ([CPUID](#))
2. Install and open CPU-Z.
3. On the **CPU** tab, record the processor **Name**.
4. On the **Memory** tab, record **Size** and **Type**.
5. Take screenshots of both tabs for your assignment.

The photo above is an **illustrative example screenshot** showing exactly where to find the required information. It shows example values of **AMD Ryzen 5 5600**, **16 GB**, and **DDR4**.

Item	Your CPU-Z reading
Processor model	i5
RAM size	16
RAM type	512

Important: Replace the example values with the readings from your actual PC; I cannot access or run CPU-Z on your computer.

2. Simulate a hardware boot failure by unplugging a component (RAM or hard drive) from a test PC, power it on, and note the BIOS/POST beep code or error message shown. Use an online POST code reference to identify the likely hardware issue and document your findings.

Hint: If your PC uses visual POST codes, note the pattern or code number displayed.

-> **Simulated Hardware Boot Failure: RAM Removed**

For a safe lab demonstration, the **RAM can be removed while the PC is completely powered off and unplugged**, then the system can be powered on briefly to observe the POST diagnostic. Do not remove or insert components while the system has power.

Procedure

1. Shut down the test PC and switch off/unplug the PSU.
2. Remove the RAM module from its DIMM slot.
3. Reconnect power and switch on the PC.
4. Observe the monitor, motherboard diagnostic LEDs, and listen for POST beeps.
5. Power the system off again before reinstalling the RAM.
6. Reinstall the RAM firmly until the retention clips lock and verify that the PC boots normally.

Example Findings

Item	Observation
Component disconnected	RAM module
Boot result	PC failed to complete normal POST
Audible/visual code	2 beeps (<i>example for supported Dell systems</i>)

Item	Observation
Online reference	Dell diagnostic/beep-code documentation
Identified problem	No memory/RAM detected
Corrective action	Reinstall/reseat the RAM module
After correction	Normal POST/boot

Dell's reference specifically identifies **beep code 2** as “**No memory or RAM detected**” on applicable systems. Dell also documents a **2-blink, pause, 3-blink** diagnostic pattern on certain OptiPlex systems for “No Memory/RAM detected.” ([Dell](#))

3. Use Memtest86 (either bootable USB or Windows version) to perform a memory diagnostic on any available PC. Record any errors found and explain what kind of real-world problems (e.g., random crashes, blue screens) these errors could cause.

-> **MemTest86 Memory Diagnostic**

I can't actually run MemTest86 on a physical PC from this chat, so the results below are an **example lab record**, not a measurement from your computer. For your actual submission, replace the example values with the results displayed by MemTest86.

Procedure

1. Download **MemTest86** and create a bootable USB drive.
2. Shut down the PC and boot it from the MemTest86 USB.
3. Start the default memory test.
4. Allow at least **one complete pass**; running several passes gives a more thorough check.
5. Record the **number of tests/passes and errors** reported.

Example Results

Test item	Result
Memory diagnostic	MemTest86
RAM installed	16 GB
Passes completed	1
Errors found	0

Test item	Result
Warnings	None
Overall result	PASS

What Memory Errors Could Cause

If MemTest86 reports memory errors, faulty or unstable RAM can lead to:

- **Random application crashes** or programs suddenly closing.
- **Windows blue-screen (BSOD) errors**, freezes, or unexpected restarts.
- **Corrupted files or failed installations**, because incorrect data can be written to storage.

4. Given this scenario: A PC randomly shuts down during gaming, but works fine on basic tasks. Using HWiNFO or GPU-Z, check and record the GPU temperature and fan speeds under load. Based on the data, suggest a possible root cause and a troubleshooting step.

-> GPU Load Diagnostic for Random Shutdown

For this scenario, I would run a game or GPU stress test for several minutes and monitor the GPU using **HWiNFO** or **GPU-Z**.

Reading	Example value under gaming load
GPU temperature	88°C
GPU fan speed	100%
GPU utilization	98–100%
Result	High temperature / maximum fan speed

Possible Root Cause

The **88°C GPU temperature with the fans already at 100%** suggests that overheating could be contributing to the shutdown. The PC may remain stable during basic tasks because the GPU produces much less heat.

Troubleshooting Step

Shut down the PC, clean dust from the GPU heatsink and case airflow paths, and verify that the GPU fans are spinning correctly. Also check that the case has adequate intake and exhaust airflow, then repeat the gaming test and compare the temperature.

Alternative: If the GPU temperature stays around 60–75°C under load but the PC still suddenly powers off, investigate the **PSU, PCIe power connection, and power delivery** rather than assuming GPU overheating.

5. Imagine you are setting up a new office for a 30-person digital marketing agency. Using online research, select hardware specifications for 2 file servers and 3 workstations (CPU, RAM, storage, GPU if needed), estimate the total budget, and justify your choices for each role.
Hint: Consider typical tasks like video editing, file sharing, and backup needs.

-> **Hardware Plan**

1. File Server 1 — Primary Shared File Server

Purpose: Central storage for campaign assets, graphics, documents, video footage, and team collaboration.

Component	Recommended specification	Estimated price
NAS/server	Synology DS925+	₹93,490
CPU	AMD Ryzen V1500B, 4-core	Included
RAM	4GB DDR4 ECC	Included
Storage	4 × 8TB Seagate IronWolf NAS HDDs	₹1,02,000
RAID	RAID 5 / Synology SHR	—
Network	2 × 2.5GbE	Included
Server 1 total		₹1,95,490

The DS925+ has a Ryzen V1500B CPU, 4GB ECC memory, four 3.5"/2.5" SATA bays, two M.2 NVMe slots and dual 2.5GbE networking. The enclosure is currently listed around ₹93,490 by an Indian retailer, while 8TB Seagate IronWolf drives are currently available around ₹25,500 each in a current Indian listing.

Why: RAID 5/SHR provides redundancy while keeping capacity useful for a shared media library. With four 8TB drives, roughly **24TB raw usable capacity** is available before filesystem overhead, while the 2.5GbE ports provide a faster network path than ordinary 1GbE.

2. File Server 2 — Backup and Disaster-Recovery Server

Purpose: Scheduled backup, snapshots, archived projects, and recovery if the primary server fails or files are accidentally deleted.

Component	Recommended specification	Estimated price
NAS/server	Synology DS925+	₹93,490
CPU	AMD Ryzen V1500B, 4-core	Included
RAM	4GB DDR4 ECC	Included
Storage	4 × 8TB Seagate IronWolf	₹1,02,000
RAID	RAID 5 / SHR	—
Network	2 × 2.5GbE	Included
Server 2 total		₹1,95,490

Why: The second unit should not simply be another live file share. I would configure it primarily for scheduled replication, versioned backups, and recovery. The second physical server protects the agency from a single-server failure; RAID alone is **not** a backup.

Both file servers total: ₹3,90,980

3. Workstation 1 — Digital Marketing / Design

Purpose: Photoshop, Illustrator, Canva, browser-heavy multitasking, social-media content, office applications, and light video work.

Component	Recommended specification	Estimated price
CPU	AMD Ryzen 5 7500F, 6-core/12-thread	₹15,095
Motherboard	B650M	₹10,300
RAM	32GB DDR5-6000	₹31,999
Storage	1TB NVMe SSD	₹16,660
GPU	GeForce RTX 4060 8GB	₹32,000
CPU cooler	DeepCool AK400-class	₹2,488

Component	Recommended specification	Estimated price
PSU	650W 80+ Bronze	₹5,199
Case	Mid-tower	₹4,000
Total		₹1,17,741

The Ryzen 5 7500F is a 6-core/12-thread AM5 CPU with a 65W default TDP. Current Indian listings show Ryzen 5 7500F around ₹15,095, B650 motherboards from about ₹10,300, 32GB DDR5-6000 kits around ₹31,999, and 1TB NVMe SSDs around ₹16,660. An ASUS RTX 4060 8GB is currently listed around ₹32,000, and its manufacturer recommends a 550W PSU, so 650W adds practical headroom.

4. Workstation 2 — Marketing / Motion-Graphics

Use the **same configuration as Workstation 1**.

Why: Standardizing the two general-purpose machines simplifies driver deployment, spare-parts management, troubleshooting, and employee replacement. The RTX 4060 provides hardware acceleration for creative applications without the cost of a high-end editing GPU.

Two general workstations total: ₹2,35,482

5. Workstation 3 — Video Editing / Content Creation

Purpose: Premiere Pro/DaVinci Resolve, 4K editing, motion graphics, color work, encoding, large project files, and GPU-accelerated effects.

Component	Recommended specification	Estimated price
CPU	AMD Ryzen 9 9900X , 12-core/24-thread	₹41,000
Motherboard	B650 ATX	₹14,900
RAM	64GB DDR5-6000	~₹64,000*
Storage	2TB PCIe 4.0 NVMe SSD	₹30,490
GPU	GeForce RTX 5070 12GB GDDR7	₹95,584
CPU cooler	DeepCool AK400 Digital Pro-class	₹4,402

Component	Recommended specification	Estimated price
PSU	750W 80+ Gold	₹7,635
Case	Airflow-focused ATX case	₹4,500
Total		~₹2,62,511

The Ryzen 9 9900X has **12 cores/24 threads**, boosts up to 5.6GHz, and supports DDR5; its current Indian price is about ₹40,191–₹41,000. The RTX 5070 has **12GB GDDR7**, while NVIDIA lists 250W total graphics power and a 650W recommended system power, making a quality 750W PSU a sensible configuration. Current Indian listings put 2TB Gen4 NVMe drives around ₹30,490 and 750W Gold PSUs around ₹7,635.

*The 64GB RAM figure is a **planning allowance** derived from current 32GB DDR5-6000 kit pricing; a matched 2×32GB kit should be purchased rather than combining unrelated kits. Current 32GB DDR5-6000 listings are around ₹31,999 and higher.

Why: Video editing benefits from more CPU cores, substantially more RAM, a stronger GPU and a larger fast NVMe workspace. NVIDIA lists AV1 encode/decode support on the RTX 5070, which is useful for modern video workflows.

Total Hardware Budget

System	Quantity	Approx. unit cost	Subtotal
Primary file server	1	₹1,95,490	₹1,95,490
Backup file server	1	₹1,95,490	₹1,95,490
General creative workstation	2	₹1,17,741	₹2,35,482
Video-editing workstation	1	~₹2,62,511	~₹2,62,511
Estimated total	5 systems		~₹8,88,973

Recommended office architecture

30 employees → 2.5GbE/managed LAN → Primary File Server → Backup File Server

The **two general workstations** can handle everyday creative/marketing production, while the **single high-performance workstation** can be reserved for demanding 4K editing, rendering, and encoding. This avoids paying high-end GPU/CPU prices for every workstation when only some workloads require them.

Budget assumption: ~₹8.9 lakh for the five core systems only. This excludes monitors, keyboards/mice, operating-system/application licenses, network switch/router upgrades, UPS units, installation labor, and external/off-site backup. Prices and stock can change quickly; several current component listings are already showing substantial variation between retailers.